

ASRA: Adaptive Scientific Reasoning Architectures for Decision Biology

Toward Intervention-Centric Scientific Intelligence Through World Models, Action Semantics Inference, and Adaptive Biological Reasoning

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Abstract

Modern artificial intelligence systems have achieved remarkable success in large-scale prediction, representation learning, and generative modeling. However, many contemporary architectures remain fundamentally limited in unfamiliar environments where objectives are partially observable, action semantics are initially unknown, causal structure must be discovered dynamically, and adaptation requires active experimentation rather than statistical interpolation over fixed training distributions.

This paper introduces ASRA (Adaptive Scientific Reasoning Architecture), a computational framework designed to investigate adaptive scientific reasoning through intervention-driven exploration, causal world-model construction, action semantics inference, abstraction-centric memory, and dynamic strategy invention. Unlike conventional systems that optimize primarily for predictive accuracy or static policy learning, ASRA reframes intelligence as an iterative scientific process involving observation, hypothesis generation, experimentation, uncertainty reduction, causal inference, and continual model refinement.

A central contribution of this work is the formalization of action semantics inference as a core computational problem for adaptive intelligence. Within ASRA, actions are initially treated as latent operators whose semantic meaning must be inferred through experimentation and transition analysis. This creates a coupled reasoning problem in which world-model construction and intervention interpretation co-evolve through active exploration.

The paper further introduces Decision Biology as the first scientific domain specialization of ASRA. Decision Biology interprets biological systems as adaptive information-processing environments operating under perturbation-driven uncertainty. Within this framework, perturbations become interventions, signaling pathways become latent biological world models, and cellular adaptation becomes sequential causal reasoning over dynamic state transitions.

Together, ASRA and Decision Biology motivate a broader computational paradigm for scientific intelligence in which AI systems are designed not merely to recognize patterns, but to autonomously reason about, experiment upon, and model complex natural systems through adaptive causal inference.

Keywords: adaptive scientific reasoning, world models, action semantics inference, Decision Biology, causal reasoning, intervention learning, biological world models, active inference, scientific intelligence, systems biology, fluid intelligence, adaptive AI, autonomous experimentation

1. Introduction

Artificial intelligence has entered an era dominated by large-scale foundation models trained on massive static datasets. These systems have demonstrated extraordinary capabilities across language generation, multimodal representation learning, protein structure prediction, and large-scale sequence modeling. Despite these advances, many contemporary architectures remain fundamentally constrained in environments requiring adaptive reasoning under uncertainty.

Most modern AI systems implicitly assume:

- stable task distributions,
- predefined objectives,
- known action semantics,
- dense supervision,
- and environments statistically similar to training data.

These assumptions enable highly effective interpolation-based learning across many domains. However, scientific environments and biological systems rarely satisfy these conditions.

Scientific discovery requires:

- hypothesis generation,
- causal experimentation,
- uncertainty-aware exploration,
- intervention analysis,
- abstraction formation,
- and iterative theory refinement.

Similarly, biological systems continuously adapt to partially observable environments through distributed signaling, perturbation-response dynamics, feedback regulation, and uncertainty-driven adaptation.

Such processes resemble adaptive scientific reasoning far more closely than static prediction.

This paper argues that future scientific intelligence systems may require computational architectures fundamentally different from conventional prediction-centric AI frameworks. In particular, adaptive intelligence may emerge not from memorization or interpolation alone, but from iterative interaction between:

- observation,
- intervention,
- causal inference,

- world-model construction,
- abstraction formation,
- and uncertainty reduction.

To investigate this hypothesis, we introduce ASRA (Adaptive Scientific Reasoning Architecture), a modular cognitive architecture designed around:

- intervention-centric exploration,
- causal world-model learning,
- action semantics inference,
- reusable abstraction memory,
- adaptive planning,
- and dynamic strategy invention.

A central technical focus of ASRA is action semantics inference: the process by which intelligent systems infer the semantic meaning of interventions through experimentation and transition analysis in environments where action dynamics are initially unknown.

The paper further introduces Decision Biology as the first scientific domain specialization of ASRA. Decision Biology interprets cellular systems as adaptive information-processing environments operating under perturbation-driven uncertainty.

Within this framework:

- perturbations become actions,
- signaling pathways become latent world models,
- cellular transitions become state dynamics,
- and biological adaptation becomes distributed causal reasoning.

This creates a conceptual bridge between:

- adaptive AI systems,
- cognitive architectures,
- information theory,
- active inference,
- causal reasoning,
- and systems biology.

The long-term objective is to motivate a broader computational paradigm for scientific intelligence systems capable of:

- autonomous hypothesis generation,
- adaptive experimentation,
- intervention planning,
- causal discovery,
- and dynamic scientific reasoning across biological and physical systems.

2. Related Work

2.1 World Models and Model-Based Reasoning

The concept of internal world models has emerged as a major direction within model-based reinforcement learning and adaptive AI systems. Prior work has explored latent predictive simulators capable of modeling environmental transitions for planning and policy optimization.

Model-based architectures such as Dreamer, MuZero, and related latent dynamics systems have demonstrated that predictive internal models can substantially improve planning efficiency and long-horizon decision-making. However, many existing world-model approaches remain primarily optimized for predictive simulation and reward maximization.

ASRA differs conceptually by emphasizing explanatory world models rather than purely predictive simulators.

The objective is not only to estimate:

$\text{State}_t + \text{Action} \rightarrow \text{State}_{t+1}$

but also to infer:

- why transitions occur,
- which latent mechanisms generate observed behavior,
- how interventions alter causal structure,
- and which experiments best reduce uncertainty.

This distinction becomes especially important for scientific environments in which causal understanding and mechanistic reasoning are central.

2.2 Active Inference and Information-Theoretic Exploration

ASRA aligns conceptually with active inference frameworks and information-theoretic approaches to adaptive behavior.

Active inference proposes that intelligent systems minimize uncertainty through adaptive interaction with the environment. Similarly, curiosity-driven exploration and Bayesian experimental design frameworks prioritize interventions that maximize information gain.

ASRA extends these ideas by integrating:

- intervention-driven exploration,
- causal hypothesis generation,
- action semantics inference,
- abstraction formation,

- and explanatory world-model construction.

Within ASRA, exploration is treated as adaptive experimental design rather than random policy exploration.

2.3 Causal Reasoning and Scientific Discovery Systems

Recent advances in causal representation learning and intervention-based reasoning have highlighted the importance of causal structure for robust generalization.

Scientific reasoning itself fundamentally depends on:

- intervention analysis,
- counterfactual reasoning,
- hypothesis testing,
- and uncertainty reduction.

ASRA reframes adaptive intelligence as a recursive scientific process operating over evolving causal models.

This differs from conventional sequence-prediction systems by emphasizing:

- dynamic theory construction,
- mechanistic inference,
- and adaptive experimentation.

2.4 Cognitive Architectures and Fluid Intelligence

Traditional cognitive architectures such as SOAR and ACT-R investigated symbolic reasoning, memory organization, and hierarchical cognition.

ASRA shares several conceptual goals with cognitive systems research, including:

- adaptive planning,
- abstraction formation,
- reusable reasoning procedures,
- and reflective evaluation.

However, ASRA differs by focusing specifically on:

- partially observable environments,
- hidden action semantics,
- intervention-driven exploration,
- and explanatory world-model learning.

2.5 Systems Biology and Perturbation Reasoning

Computational biology has increasingly shifted toward dynamical systems modeling, perturbation-response analysis, and signaling-network reasoning.

Biological systems continuously process information through:

- signaling pathways,
- feedback regulation,
- adaptive state transitions,
- and distributed causal interactions.

Recent advances in perturbation biology, single-cell sequencing, and large-scale perturbation-response datasets have enabled increasingly sophisticated models of cellular adaptation.

Decision Biology extends this trajectory by reframing biological systems as adaptive reasoning environments operating under uncertainty.

Within this framework:

- perturbations become interventions,
- signaling pathways become world models,
- and cellular adaptation becomes sequential causal reasoning.

3. From Prediction-Centric AI to Adaptive Scientific Reasoning

A central limitation of many contemporary AI systems is their dependence on interpolation across fixed training distributions.

Although such systems excel at:

- large-scale prediction,
- retrieval,
- representation learning,
- and sequence completion,

adaptive scientific reasoning requires additional capabilities.

An intelligent system operating in an unfamiliar environment must:

1. infer hidden mechanics,
2. identify latent variables,
3. discover causal relationships,
4. generate hypotheses,
5. design experiments,
6. revise internal theories,
7. compress observations into reusable abstractions,
8. and dynamically invent new strategies.

Scientific reasoning itself operates through this recursive adaptive loop.

ASRA therefore models intelligence as an iterative process of:

- observation,
- intervention,
- causal inference,
- abstraction formation,
- and world-model refinement.

Rather than optimizing directly for static output prediction, the architecture attempts to construct explanatory internal models capable of supporting:

- adaptive exploration,
- intervention planning,
- counterfactual simulation,
- and mechanistic reasoning.

This creates a shift from:

prediction-centric intelligence

toward:

intervention-centric scientific reasoning.

4. ASRA: Adaptive Scientific Reasoning Architecture

4.1 Architectural Philosophy

ASRA is a modular computational architecture designed to investigate fluid intelligence in partially observable environments with unknown mechanics.

The architecture is built around several principles:

- adaptive experimentation,
- active inference,
- causal modeling,
- world-model construction,
- uncertainty-aware planning,
- reusable abstraction learning,
- and dynamic strategy invention.

At the center of ASRA is a recursive scientific reasoning loop:

Observe → Hypothesize → Experiment → Analyze → Refine Theory → Adapt → Retry

The architecture behaves less like a static predictor and more like a scientific investigator attempting to understand environmental dynamics through intervention and experimentation.

4.2 Observation and Representation Layer

The observation system transforms raw environmental states into structured internal representations.

Responsibilities include:

- object extraction,
- spatial abstraction,
- topology analysis,
- connected-component detection,
- relational graph construction,
- symmetry analysis,
- and temporal transition tracking.

The objective is not merely dimensionality reduction, but semantic representation.

The architecture attempts to identify meaningful latent structures that downstream reasoning systems can operate upon.

4.3 Hypothesis Generation Engine

ASRA continuously generates candidate explanations for observed environmental behavior.

Hypotheses may involve:

- causal rules,
- action semantics,
- transition dynamics,
- hidden constraints,
- latent objectives,
- or object interaction laws.

Unlike static symbolic systems, hypotheses remain probabilistic and continuously updated through experimentation.

The system therefore behaves less like a fixed program and more like an evolving scientific model-construction process.

4.4 Experimental Simulation and Intervention Layer

One of ASRA's defining properties is its emphasis on experimentation.

The architecture actively performs interventions to reduce uncertainty and separate competing hypotheses.

This component evaluates:

- intervention effects,
- counterfactual trajectories,
- causal consistency,
- uncertainty reduction,
- and information gain.

Exploration therefore becomes hypothesis-driven experimentation rather than random action selection.

4.5 Meta-Abstraction Memory

Conventional systems frequently store:

- memorized mappings,
- benchmark-specific solutions,
- or narrow optimization heuristics.

ASRA instead stores reusable conceptual abstractions.

These include:

- causal motifs,
- exploration procedures,
- reasoning operators,
- structural schemas,
- failure patterns,
- and transferable conceptual strategies.

The objective is conceptual reuse rather than memorization.

This distinction is critical for fluid intelligence and open-ended adaptation.

4.6 Strategy Invention

A central goal of ASRA is dynamic strategy invention.

When existing abstractions fail, the architecture attempts to generate novel reasoning procedures.

Examples include:

- hierarchical decomposition,
- topology-preserving transformations,
- recursive planning,
- causal graph traversal,
- and object-centric reasoning strategies.

This moves the architecture beyond retrieval-based intelligence toward constructive intelligence.

5. Action Semantics Inference

5.1 The Action Semantics Problem

One of the central technical problems addressed by ASRA is action semantics inference.

Traditional AI systems typically assume:

- predefined action spaces,
- stable transition dynamics,
- and known intervention semantics.

However, adaptive scientific environments rarely expose these assumptions explicitly.

In many real-world systems, intelligent agents must infer:

- what actions do,
- when actions are valid,
- which variables are causally affected,
- how interventions alter environmental dynamics,
- and which hidden mechanisms generate observed transitions.

ASRA therefore treats actions as latent operators whose semantics must be inferred through experimentation.

5.2 Formalization

The action semantics inference problem may be formalized as follows.

Let the environment be represented as:

$$E = (S, A, T, O)$$

where:

- S represents latent environmental states,
- A represents interventions/actions,
- T represents unknown transition dynamics,
- O represents observations.

Unlike conventional reinforcement learning frameworks, ASRA assumes that the semantic interpretation of actions is initially unknown.

Traditional reinforcement learning systems typically assume:

- ACTION1 = move left
- ACTION2 = jump
- ACTION3 = rotate object

In such systems, the meaning of actions is predefined.

ASRA instead assumes that action semantics are initially latent and must be inferred through experimentation.

The system only observes:

state_t → action → next_state

and must infer:

“What does this action actually do?”

This creates the core action semantics inference problem.

Each action $a \in A$ is therefore treated as a latent operator whose semantic meaning must be inferred from observed state transitions.

The architecture attempts to estimate:

$$\hat{\phi}(a_t) = \arg\max_{\phi} P(s_{t+1} \mid s_t, \phi(a_t), \theta)$$

where:

- a_t represents the intervention/action at time t ,
- $\phi(a_t)$ represents a hypothesized semantic interpretation of the action,
- $\hat{\phi}(a_t)$ represents the inferred semantic meaning,
- s_t represents the current environmental state,
- s_{t+1} represents the observed next state,
- θ represents latent environmental mechanics,

- and $P(s_{t+1} \mid s_t, \phi(a_t), \theta)$ represents the probability of observing the next state given the current state, hypothesized action semantics, and hidden environmental dynamics.

The operator:

argmax_{ϕ}

selects the semantic interpretation that best explains the observed transition.

In plain terms, the architecture evaluates competing hypotheses for what an action means and selects the interpretation that most consistently explains observed environmental behavior.

For example, suppose the system observes:

- state_t: agent at position (5,5)
- action: ACTION2
- next_state: agent at position (4,5)

The system may generate competing hypotheses:

- ACTION2 = move upward
- ACTION2 = rotate object
- ACTION2 = activate local transformation

The architecture then evaluates which interpretation most plausibly explains the observed transition.

If repeated experimentation consistently shows upward movement after ACTION2, the system progressively converges toward:

$\hat{\phi}(\text{ACTION2}) = \text{MOVE_UP}$

Action semantics inference therefore becomes the process of constructing causal semantic mappings between interventions and environmental dynamics through iterative experimentation, uncertainty reduction, and transition analysis.

This formulation is important because many real-world scientific environments do not expose intervention semantics explicitly.

Scientists themselves frequently operate under similar conditions:

- perturbation effects are partially unknown,
- causal dynamics must be inferred experimentally,
- and intervention meaning emerges through observation and hypothesis refinement.

ASRA therefore treats action understanding as a scientific inference problem rather than a predefined control interface.

5.3 Transition-Centric Inference

The architecture continuously evaluates:

- state-transition differences,
- intervention outcomes,
- counterfactual trajectories,
- uncertainty reduction,
- and causal consistency.

The objective is not merely policy optimization.

Instead, the system attempts to infer the underlying mechanisms governing environmental behavior.

For example, when an intervention produces a structural transformation in environmental state, ASRA attempts to determine whether the transition represents:

- movement,
- topological transformation,
- object interaction,
- state propagation,
- constraint activation,
- or latent-rule modification.

Action semantics therefore emerge through iterative causal experimentation.

5.4 Information-Theoretic Action Discovery

ASRA approaches action discovery through information-theoretic reasoning.

The architecture prioritizes interventions that maximize information gain about competing hypotheses.

Intervention selection may be formalized as:

$$a_t^* = \arg\max_{a_t} I(H_t; s_{t+1} \mid a_t)$$

where:

- H_t represents the current hypothesis distribution,
- I represents mutual information.

The system continuously asks:

- Which intervention most reduces uncertainty?
- Which action best separates competing hypotheses?
- Which experiment most efficiently improves the world model?

- Which transition reveals hidden causal structure?

This transforms exploration into adaptive experimental design.

5.5 Counterfactual Reasoning

Action semantics inference is tightly coupled with counterfactual reasoning.

The architecture attempts to evaluate:

- what would have occurred under alternative interventions,
- which transitions are causally attributable to specific actions,
- and how environmental trajectories diverge under competing interventions.

Counterfactual simulation enables:

- causal disambiguation,
- intervention planning,
- uncertainty-aware exploration,
- and dynamic strategy refinement.

6. World Models and Explanatory Scientific Reasoning

6.1 Explanatory vs Predictive World Models

ASRA continuously constructs internal world models representing environmental dynamics.

The architecture attempts to estimate:

$$\mathbf{g}_{t+1} = \mathbf{g}_t + \mathbf{M}_t : (\mathbf{s}_t, \mathbf{a}_t, \mathbf{h}_t) \rightarrow \hat{\mathbf{s}}_{t+1}$$

where:

- $\mathbf{M}_t$ represents the evolving world model,
- $\mathbf{h}_t$ represents latent hypotheses.

Importantly, ASRA does not treat world models merely as predictive simulators.

The architecture instead attempts to construct explanatory models capable of supporting:

- causal interpretation,
- intervention analysis,

- counterfactual reasoning,
- and uncertainty-aware experimentation.

Predictive systems ask:

“What will happen next?”

ASRA additionally asks:

- “Why did this transition occur?”
- “Which hidden mechanism generated this effect?”
- “Which intervention best separates competing explanations?”

This distinction is essential for scientific reasoning.

6.2 Coupled World-Model and Action-Semantics Learning

Action semantics inference and world-model construction are deeply coupled.

A world model cannot become predictive unless intervention semantics are understood.

Simultaneously, intervention semantics cannot be inferred without a partially coherent model of environmental dynamics.

ASRA therefore jointly updates:

- semantic interpretations of interventions,
- causal transition dynamics,
- latent environmental structure,
- and explanatory hypotheses.

As experiments accumulate, the architecture constructs progressively more stable causal representations of environmental behavior.

6.3 Scientific World Models

Within ASRA, world models function as adaptive scientific theories.

These models attempt to represent:

- environmental laws,
- causal interactions,
- latent constraints,
- state-transition dynamics,
- and intervention consequences.

The architecture therefore reframes intelligence as dynamic scientific model construction rather than static statistical interpolation.

7. Information-Theoretic Foundations

ASRA strongly aligns with information-theoretic reasoning.

The architecture treats exploration as an information acquisition process.

Core principles include:

- entropy reduction,
- uncertainty minimization,
- mutual information maximization,
- causal compression,
- and adaptive model refinement.

Within this framing, intelligent exploration is not brute-force search.

It is targeted uncertainty reduction.

Intervention selection may also be expressed as:

$$a^* = \arg\max_a \left[H(M_t) - H(M_{t+1} \mid a) \right]$$

where:

- $H(M_t)$ represents uncertainty in the current world model.

The system therefore continuously seeks experiments that most efficiently improve causal understanding.

This creates a conceptual bridge between:

- adaptive AI,
 - scientific experimentation,
 - active inference,
 - and biological information processing.
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8. Decision Biology

8.1 Biological Systems as Adaptive Reasoning Environments

Decision Biology represents the first scientific domain specialization of ASRA.

The central hypothesis of Decision Biology is that biological systems can be interpreted as adaptive information-processing environments operating under perturbation-driven uncertainty.

Within this framework:

- cells continuously receive signals,
- infer environmental conditions,
- process uncertainty,
- regulate internal states,
- coordinate through communication,
- and adapt through distributed causal interactions.

This creates a structural analogy between adaptive reasoning systems and biological systems.

8.2 Perturbations as Interventions

Within Decision Biology, perturbations become biological interventions.

Examples include:

- drug exposure,
- CRISPR perturbation,
- ligand binding,
- signaling activation,
- metabolic stress,
- and environmental change.

Cells do not passively react to these perturbations.

Instead, biological systems dynamically infer, propagate, and adapt to intervention effects through distributed signaling networks.

This creates a direct conceptual mapping:

ASRA	Decision Biology
Action	Perturbation
World model	Signaling network

ASRA	Decision Biology
State transition	Cellular adaptation
Intervention analysis	Biological experimentation
Hypothesis generation	Mechanistic pathway inference
Counterfactual simulation	Predictive pathway reasoning

8.3 Biological World Models

Within Decision Biology, signaling systems become latent biological world models.

Biological dynamics may be represented as:

$$x_{t+1} = F(x_t, p_t, \omega_t)$$

where:

- x_t represents cellular state,
- p_t represents perturbation,
- ω_t represents latent regulatory dynamics.

The architecture attempts to infer:

cell state + perturbation $\rightarrow$ next cell state

through adaptive intervention-driven reasoning.

These biological world models attempt to represent:

- signaling pathways,
- regulatory interactions,
- perturbation-response dynamics,
- adaptive state trajectories,
- and latent biological mechanisms.

8.4 Perturbation Semantics Inference

One of the most important implications of Decision Biology is that biological perturbations themselves possess latent semantics.

A perturbation may:

- inhibit signaling propagation,
- activate stress-response pathways,

- induce compensatory adaptation,
- alter metabolic regulation,
- or trigger latent state transitions.

The semantic meaning of perturbations must therefore be inferred from distributed downstream responses.

This creates a biological analogue of action semantics inference.

Decision Biology therefore extends ASRA by treating biological systems as partially observable adaptive reasoning environments whose latent dynamics must be inferred through intervention and observation.

9. Adaptive Scientific Intelligence

ASRA and Decision Biology together motivate a broader computational paradigm for scientific intelligence.

The long-term objective is not merely improved prediction.

Instead, the broader goal is the development of adaptive scientific intelligence systems capable of:

- autonomous hypothesis generation,
- adaptive experimentation,
- causal reasoning,
- abstraction transfer,
- intervention planning,
- uncertainty-aware exploration,
- and multi-scale scientific reasoning.

Such systems may eventually support:

- biological discovery,
- therapeutic design,
- synthetic biology,
- environmental systems modeling,
- autonomous laboratory systems,
- and adaptive scientific research.

Within this hierarchy:

- ASRA functions as the adaptive reasoning engine,
 - Decision Biology functions as the first scientific application domain,
 - and broader scientific world-model systems may eventually evolve into Nature Foundation Models.
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10. Limitations

Several limitations should be acknowledged.

First, ASRA currently represents a theoretical and computational framework rather than a fully validated experimental system.

Although the architecture proposes mechanisms for:

- action semantics inference,
- world-model construction,
- and adaptive experimentation,

large-scale empirical validation remains future work.

Second, biological systems possess enormous dynamical complexity across molecular, cellular, tissue, and organism scales.

Accurate biological world models may require:

- multi-scale integration,
- causal graph inference,
- temporal modeling,
- and large-scale perturbation-response datasets.

Third, computational costs associated with adaptive experimentation and large-scale causal simulation may become substantial in highly complex environments.

Finally, many components of scientific reasoning remain incompletely understood from both computational and cognitive perspectives.

ASRA therefore should be viewed as a research framework for investigating adaptive scientific reasoning rather than a complete solution to general intelligence.

11. Future Directions

Several future research directions emerge from this framework.

11.1 Adaptive Perturbation Reasoning

Future systems may perform uncertainty-aware biological intervention planning through adaptive perturbation experimentation.

11.2 Multi-Scale Scientific Modeling

Future architectures may integrate:

- molecular,
- cellular,
- tissue,
- organism,
- and environmental dynamics

within unified world-model systems.

11.3 Autonomous Hypothesis Generation

Scientific intelligence systems may eventually generate, prioritize, and experimentally evaluate novel scientific hypotheses.

11.4 Biological Abstraction Learning

Future systems may learn reusable biological abstractions transferable across domains and organisms.

11.5 Scientific Agent Architectures

Multi-agent scientific reasoning systems may specialize in:

- experimentation,
- modeling,
- validation,
- causal analysis,
- and theory refinement.

11.6 Decision Biology Datasets

Several emerging datasets strongly align with the Decision Biology framework, including:

- LINCS L1000,
- scPerturb,
- Cell Painting,
- OmniPath,
- Human Cell Atlas,
- and perturbation-response transcriptomic datasets.

These datasets naturally support:

- perturbation semantics inference,
- causal biological reasoning,

- latent-state discovery,
 - signaling-network modeling,
 - and adaptive response prediction.
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12. Conclusion

ASRA proposes a shift from prediction-centric AI toward adaptive scientific reasoning architectures.

Rather than treating intelligence as memorization or statistical interpolation, the framework views intelligence as:

- dynamic world-model construction,
- intervention-centric reasoning,
- abstraction formation,
- experimentation,
- causal inference,
- and adaptive strategy invention.

A central contribution of this work is the formalization of action semantics inference as a core computational problem for adaptive intelligence.

Within ASRA, interventions are treated as latent operators whose semantics must be inferred through experimentation and causal transition analysis.

Decision Biology extends this framework into biological systems by reframing cells as adaptive information-processing systems operating under perturbation-driven uncertainty.

Together, ASRA and Decision Biology motivate a broader computational paradigm for scientific intelligence systems capable not merely of recognizing patterns, but of reasoning about complex natural systems through experimentation, intervention, abstraction, and adaptive model construction.

The long-term vision is the emergence of AI systems that behave less like static predictors and more like adaptive scientific investigators capable of understanding, modeling, and interacting with the causal structure of nature.

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